

Sampling MPC and heuristic advisors on ARLAMX V2.0

Single-advisor AstroChimera session. The plant is the V2.0 C++ library (GGM03S degree 4, Sentman M-01, panel SRP, co-rotating MSIS 2.1). The spacecraft has no sun sensor. Heuristics close only on generated power. The MPC may use a sun ephemeris (clock + orbit), which is not a sun sensor.

Geometry for every number: SolarCat 1 % sealed-side mesh, 0.625 kg, $i = 23^\circ$, $e = 0.001$, $a = R_E + 500$ km, $F_{10.7} = 150$, $A_p = 4$.

1. What the advisors adjust

Both families emit one scalar-first unit quaternion q_{BN} ($q_0 \geq 0$) every 300 s advisor step. The plant tracks it with MRP-PD at 2 s. That quaternion is the only command. Body +Z is simultaneously the sail normal, the two-sided panel axis, and the antenna boresight. There is no array gimbal.

2. Heuristic family

2.1 SoC mode ladder

Mode selection uses only supercapacitor state of charge (Wertz 1978). Default thresholds $t = (0.60, 0.50, 0.40, 0.30, 0.20)$.

SoC	Command
$\geq t_0$	+Z at a visible station, else min-drag / nadir / last-good sun
$t_1 \dots t_0$	station if visible, else last-good sun, else search
$< t_1$	power search; no station

Search steps grow as SoC falls ($\times 1.5$ below t_2 , $\times 2$ below t_3). Below t_4 the search latches on any sensed power.

2.2 Power search without a sun vector

Sweep about body Y then X until $\text{power_gen_norm} \geq \text{sense_floor}$. Then coordinate-descent hill-climb on $\pm X$, $\pm Y$ (Nocedal & Wright 2006): keep a step if generation rose by more than 10^{-3} , else revert; halve the step after a failed cycle. Hold when generation $\geq \text{gen_target}$ or the step is < 1 deg. A fruitless 360° sweep on both axes is treated as eclipse: freeze for $\text{eclipse_hold_steps}$ (default 6 = 30 min). Body-frame updates are $q' = q_d \times q_{BN}$ (Hamilton product, left multiplication, passive convention).

3. Sampling MPC

The attitude–power–drag map is non-convex and Sentman FMF has no cheap derivative. A receding-horizon sampler is used (Rawlings, Mayne & Diehl 2017; Camacho & Bordons 2007).

3.1 Candidates and the 40° clip

Each step proposes: hold, sun-track from the analytic ephemeris (optional), ground-station track if elevation ≥ 10 deg, min-drag, nadir, and $K = 8$ random axis–angle perturbations. Every candidate is SLERP-clipped to 40° from the incumbent:

$$q(\tau) = [\sin((1-\tau)\Omega/2) q_a + \sin(\tau\Omega/2) q_b] / \sin(\Omega/2), \Omega = 2 \arccos |< q_a, q_b >|, \tau = \min(1, 40^\circ/\Omega).$$

3.2 Surrogate (H = 6 steps = 30 min)

The candidate is held inertially, matching the plant's zero-order hold. Orbit: two-body RK4, 60 s, $\mu = 3.986004418 \times 10^{14} \text{ m}^3/\text{s}^2$ (Vallado 2013 §1). $J_2/\text{drag}/\text{SRP}$ are omitted — they do not change ranking over 30 min. Eclipse: cylindrical shadow. Power: two-sided cosine $P_{\text{gen}} = 0.78 \times 0.85 \times |z_N \cdot s| \times 1[\text{lit}]$. Loads: 7 mW always, 205 mW GPS when lit, 400 mW TX if $z \cdot g > 0.7$ and lit. Supercap 0.53 Wh. Drag: Sentman coefficients_only on the real 1 % panels.

3.3 Objective

$$J(q) = \sum_k [B(\text{SoC}_k) + 1[\text{pass, lit}](T(\theta_{gs}) + w_{\text{align}} \cos \theta_{gs}) - w_{\text{cd}} C_{d,k}] - w_{\text{slew}} \Delta_{\text{cmd}}/40^\circ.$$

$B(s) = -4(0.4-s)/0.4$ if $s < 0.4$; $+1$ if $0.4 \leq s \leq 0.6$; $\exp(-6(s-0.6))$ if $s > 0.6$. T is the $2^\circ/5^\circ/10^\circ$ tier ladder (full / half / quarter / -0.5).
Weights: power 1, low-SoC 4, GS 2, GS penalty 0.5, align 1, Cd 0.5, slew 0.2.

The physical plant still runs GGM03S + Sentman + SRP + MSIS. The surrogate lives only inside the planner. Variant `mpc_no_sun_eph` drops the ephemeris sun-track candidate so the planner has the same solar information as the heuristic.

4. Eclipse and ground-pointing limits

Two-sided $\pm Z$ panels make nadir a **power-positive** communications attitude: at local noon r is nearly sunward, so the anti-sun face still illuminates and $|z \cdot s| \sim 1$. The 2-day run measured mean sunlit generation 0.81 of peak at nadir. Tilting $0^\circ/15^\circ/30^\circ/45^\circ/60^\circ$ toward velocity keeps min SoC at 0.528/0.523/0.517/0.511/0.504 and generation at 0.81/0.79/0.77/0.74/0.71 — a 60° ground-looking tilt is still eclipse-stable. One eclipse (~ 35 min) at the GPS+sensing load drops about 0.23 SoC. Heuristics that latch a dim false sun (sense_floor 0.02, or hold-on-any-power) brown out; floors 0.05–0.20 with a real climb target survive. Stations for $i = 23^\circ$: Honolulu, Mexico City, Mumbai, Singapore, Quito, Darwin.

5. Lifetime context (min / max drag)

Prescribed-attitude decay on the same 1 % plant, $F_{10.7}=150$, $A_p=4$, corotating + SRP: min-drag reaches 250 km in **18.81 days**, max-drag in **10.85 days**. Start $|r|-R_E = 493.12$ km is perigee of $e = 0.001$, not a first-step drop.

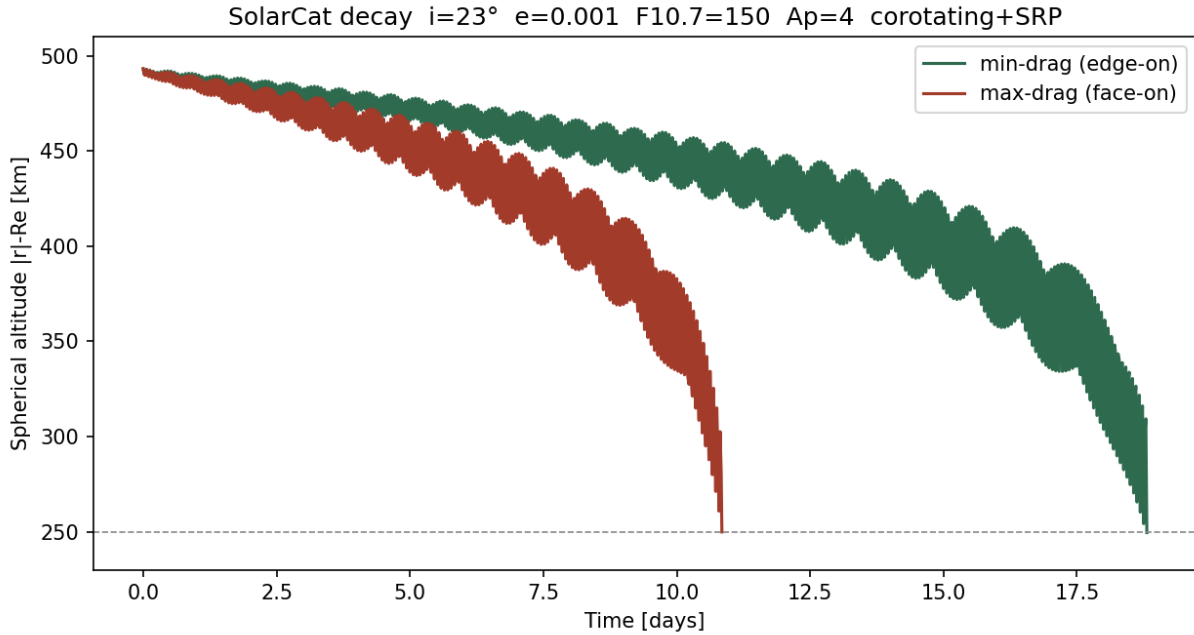


Figure 1. Min- and max-drag altitude. Oscillation is the eccentric + J_2 radius, growing as drag pumps e .

6. Two-day advisor campaign (1 % SolarCat)

Policy	min SoC	mean SoC	brown	gen (lit)	GS lock	nadir 30°	dh km	ok
mission_v17	0.322	0.937	0	0.472	0.049	0.007	-13.39	yes
conserve	0.000	0.016	542	0.189	0.023	0.000	-12.58	no
eclipse_hold	0.362	0.946	0	0.474	0.049	0.005	-13.39	yes
eclipse_deep	0.322	0.941	0	0.472	0.047	0.010	-13.55	yes
power_first	0.382	0.954	0	0.472	0.045	0.009	-13.29	yes
ground_first	0.528	0.985	0	0.802	0.052	0.247	-16.51	yes
ground_rich_nadir	0.322	0.963	0	0.786	0.054	0.238	-16.38	yes
low_threshold_search	0.483	0.951	0	0.473	0.047	0.005	-13.31	yes
fine_search	0.321	0.925	0	0.470	0.050	0.005	-13.38	yes
coarse_search	0.410	0.862	0	0.441	0.043	0.014	-13.16	yes
high_floor	0.262	0.934	0	0.470	0.049	0.009	-13.41	yes
any_power_hold	0.000	0.021	527	0.195	0.035	0.000	-10.58	no
search_only	0.325	0.923	0	0.418	0.030	0.009	-12.65	yes
min_drag	0.425	0.930	0	0.449	0.035	0.000	-12.46	yes
max_drag	0.489	0.982	0	0.659	0.014	0.073	-21.13	yes
nadir	0.528	0.987	0	0.811	0.028	0.252	-16.25	yes
ground_station	0.528	0.985	0	0.802	0.052	0.247	-16.51	yes
off_nadir_0	0.528	0.987	0	0.811	0.028	0.252	-16.25	yes
off_nadir_15	0.523	0.987	0	0.792	0.021	0.252	-17.43	yes
off_nadir_30	0.517	0.986	0	0.767	0.021	0.229	-18.61	yes
off_nadir_45	0.511	0.985	0	0.741	0.016	0.198	-19.67	yes
off_nadir_60	0.504	0.985	0	0.713	0.016	0.161	-20.49	yes
search_f02_t30	0.000	0.044	452	0.306	0.036	0.000	-12.27	no
search_f05_t50	0.325	0.914	0	0.421	0.035	0.000	-12.49	yes
search_f10_t70	0.289	0.934	0	0.423	0.033	0.000	-13.00	yes
search_f20_t85	0.375	0.897	0	0.402	0.031	0.014	-11.39	yes
mpc	0.361	0.417	0	0.336	0.021	0.002	-7.75	yes
mpc_no_sun_eph	0.370	0.415	0	0.335	0.010	0.000	-7.49	yes

Table 1. Two-day closed loop, 300 s advisor, 2 s MRP-PD. GS lock is the fraction of steps with a visible station and $z \cdot g > 0.7$. Nadir 30° is the fraction of steps with +Z within 30° of -r. dh is spherical-altitude change over the two days (more negative = more drag).

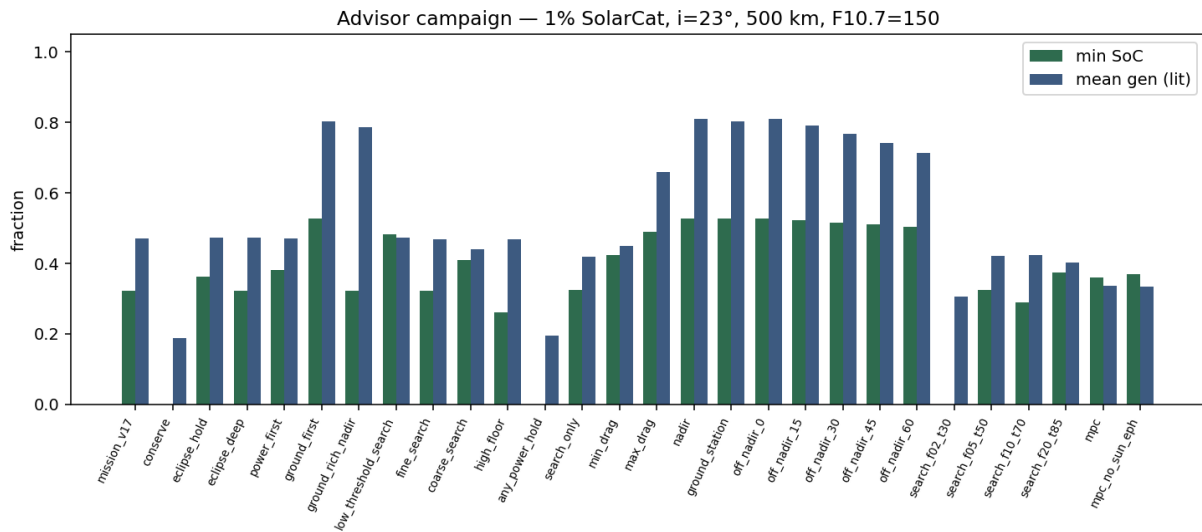


Figure 2. Minimum SoC and mean sunlit generation by policy.

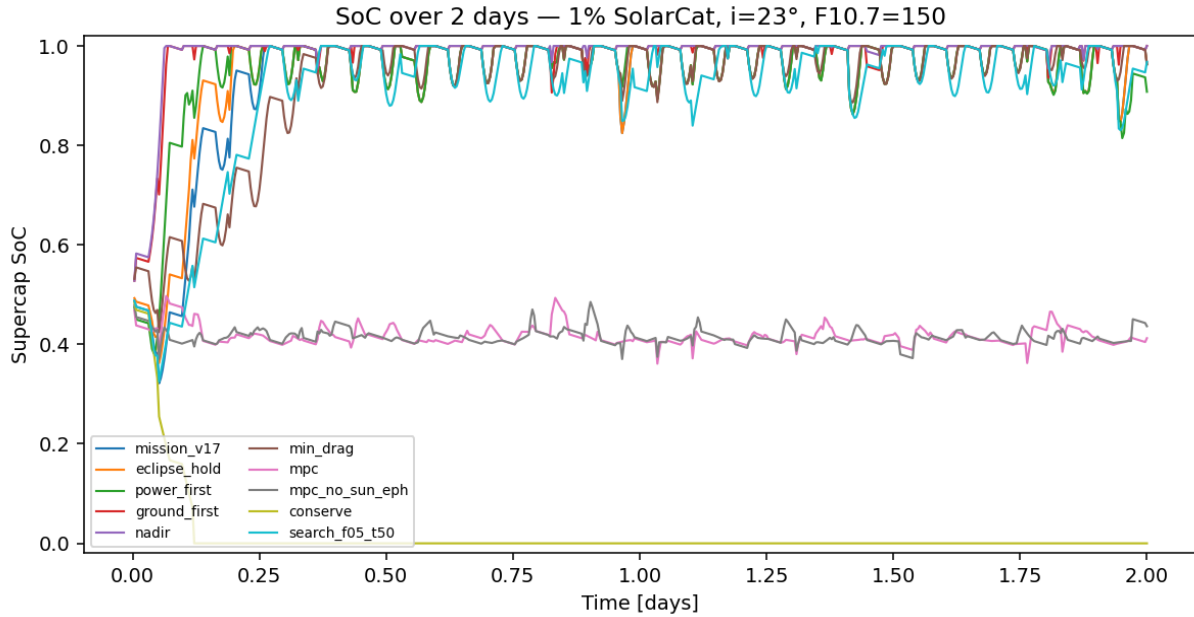


Figure 3. SoC time history. conserve and the 0.02-floor search go to zero; nadir / ground_first stay near 0.53.

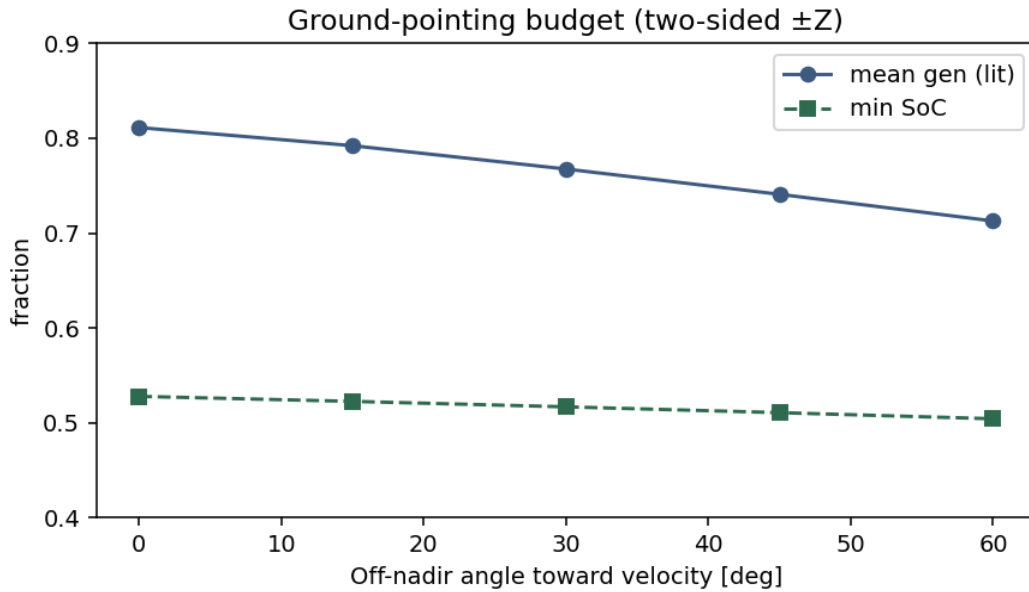


Figure 4. Off-nadir tilt toward velocity. Generation and min SoC stay above 0.50 out to 60°.

6.1 How to read the table

A policy that **survives eclipse** has min SoC well above 0 and zero brownouts. On this bus that means entering umbra above ~ 0.35 SoC and generating at least 0.4 of peak when lit. **Ground pointing** shows up as a high nadir-30° fraction and/or a non-zero GS lock; both cost generation because +Z is then off the sun. The search-floor rows isolate how bright the panels must be before the heuristic believes it has found the sun — without ever reading a sun vector.

7. Citations

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